بسم الله الرحمن الرحيم

Prof. Moha

Amino Acids Metabolism

الأستاذ الدكتور

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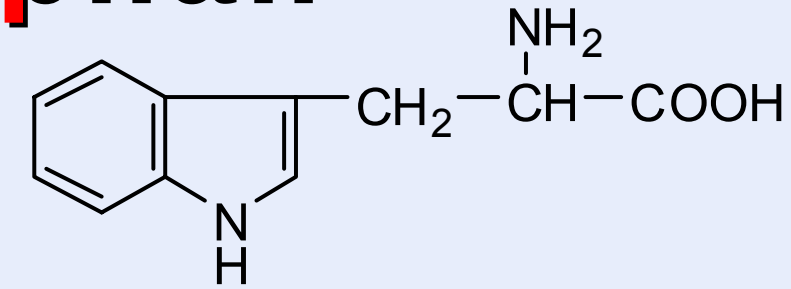
Lecture 8

- **Tryptophan**

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Tryptophan



Structure

- It is α-amino, β-indole propionic

Nutrional Value

- It is essential amino acid

Metabolic Fate

- It is both glucogenic and ketogenic

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Tryptophan

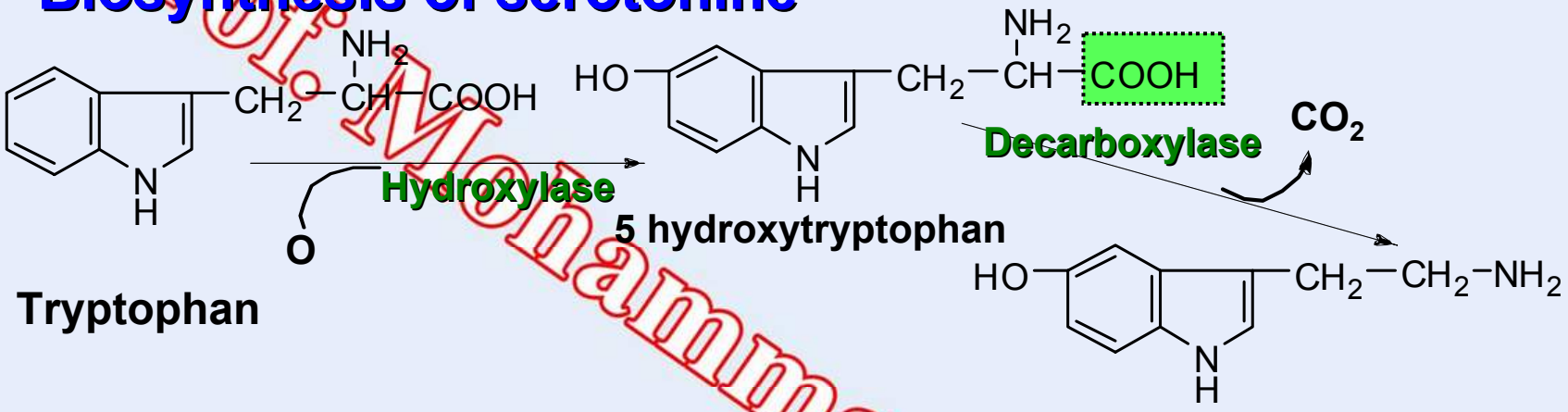
Metabolic Fate

1. Incorporated in protein structure
2. Biosynthesis of seretonine
3. Biosynthesis of melatonin
4. Biosynthesis of Niacin
5. Synthesis of glucose
6. Synthesis of ketone bodies
7. Putrefaction

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Tryptophan

Biosynthesis of serotonine



Tryptophan

5 hydroxytryptophan

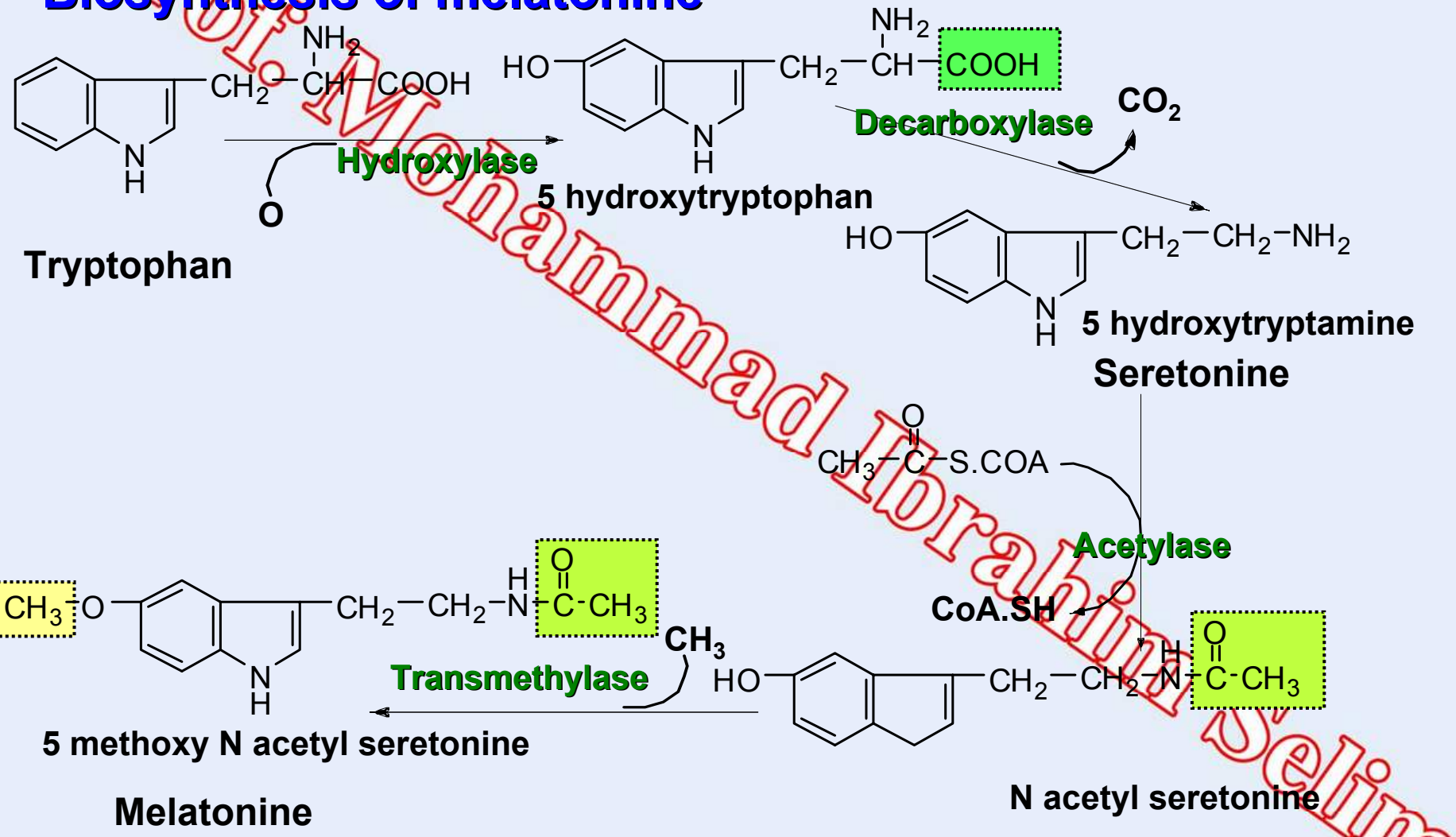
5 hydroxytryptamine

Seretonine

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Tryptophan

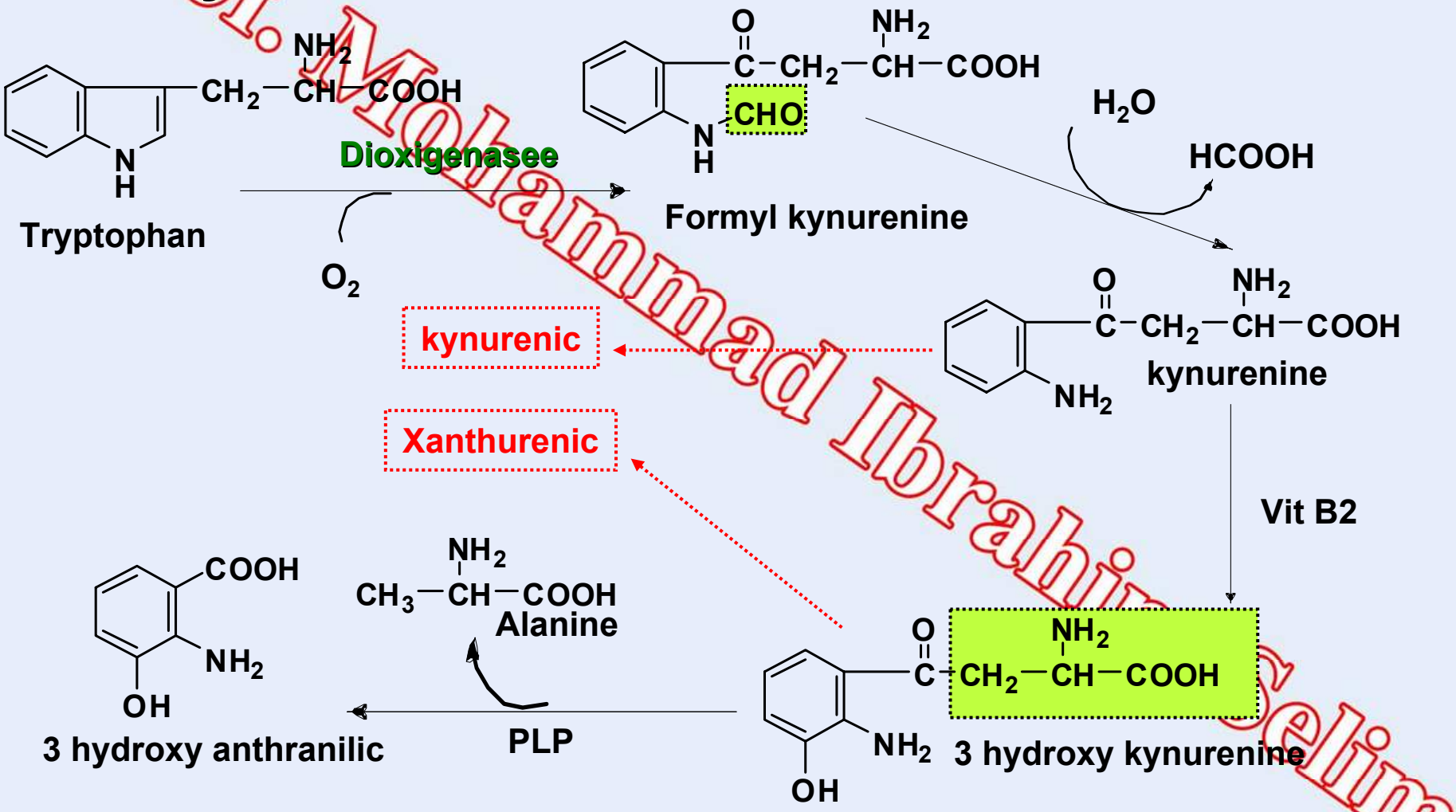
Biosynthesis of melatonine



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Tryptophan

Biosynthesis of niacin

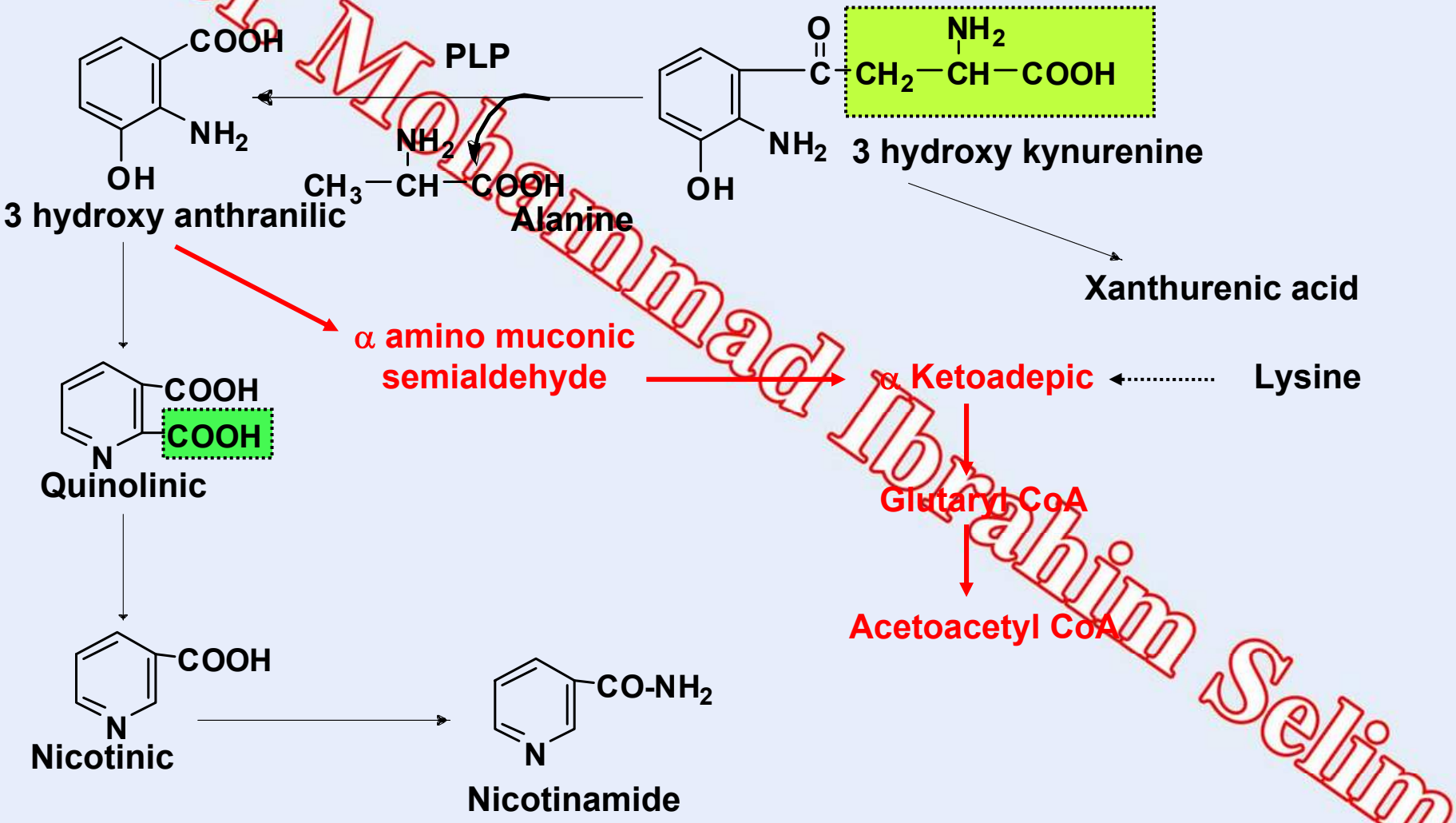


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Tryptophan

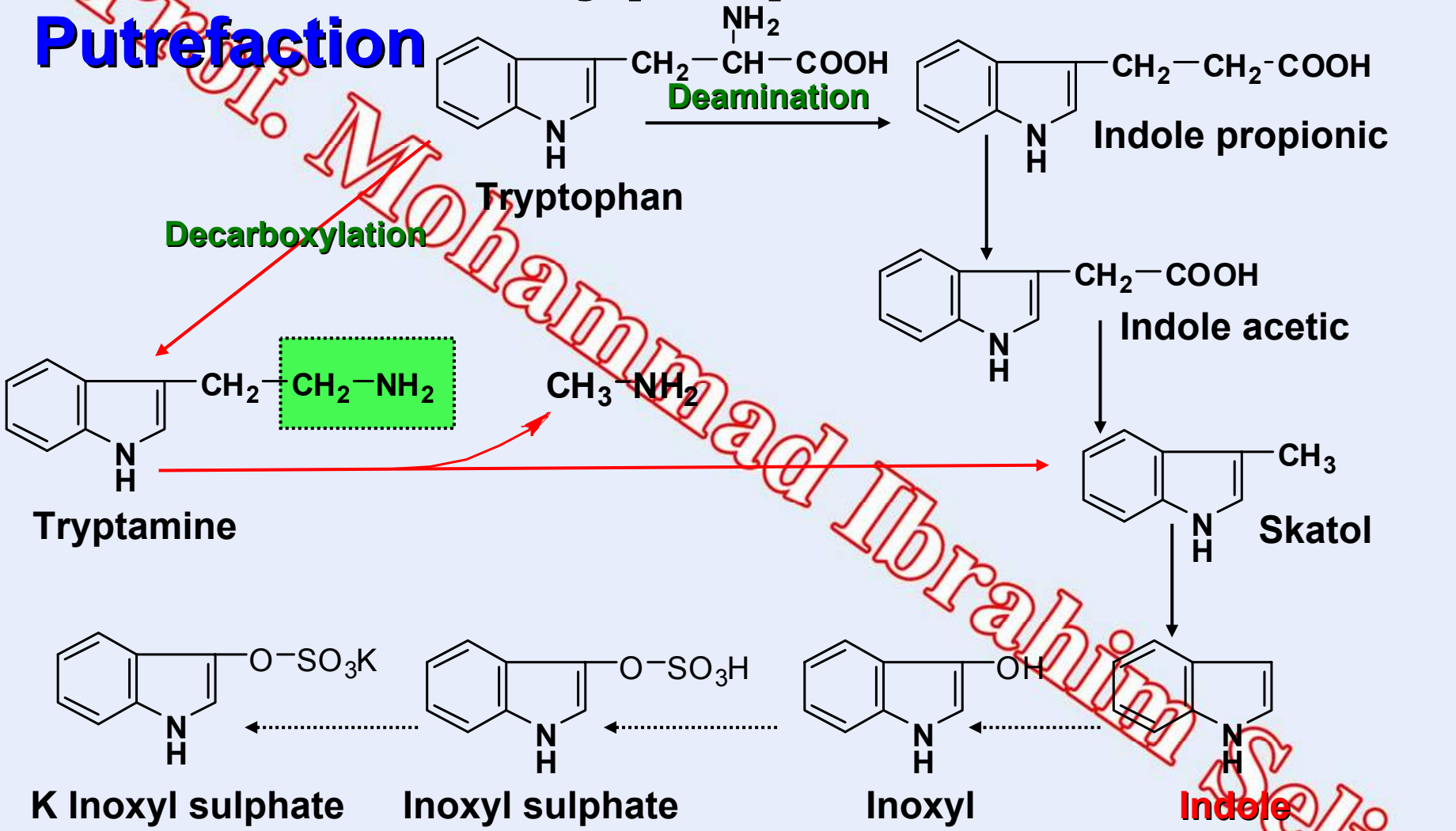
Biosynthesis of niacin



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Tryptophan

Putrefaction



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Tryptophan

Inborn error of Tryptophan metabolism

- **Hartnup disease**
- **Cause**
  - Defect in intestinal and renal transport of:
    - **Tryptophan**
    - **Other neutral amino acids**
- **Diagnosis**
  - Pellagra like symptoms
  - Psychological abnormalities
  - Mental retardation
  - Neutral amino aciduria
  - Putrefactive products in urine (Indican)
- **Treatment**
  - Vitamin B3 (Nicotinic acid)

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